

M19 Learner Book

Interview, Technical English & Job Launch | Release Candidate 1 | 14 controlled lessons

M19-L01 - Truthful one-page Data Engineer resume

Learning objective

Produce a truthful, evidence-backed job artifact or targeting decision.

Why this matters in Data Engineering

Job-launch quality depends on defensible evidence and targeting, not inflated claims.

Understand

- Every claim must map to course/project evidence.
- Preferred requirements are not automatically hard filters; explicit must-have constraints matter.
- Keep qualification, employment, location and experience statements factual.

Try / inspect

evidence -> claim -> verification -> publish

Common failure

Inflating experience/title, copying generic text or applying blindly without tracking reasons.

Your turn - independent proof

Create a one-page resume; map every bullet to evidence before removing internal tags.

Move on when

The artifact/decision can be defended line by line and revised from evidence.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L02 - GitHub and project portfolio final polish

Learning objective

Produce a truthful, evidence-backed job artifact or targeting decision.

Why this matters in Data Engineering

Job-launch quality depends on defensible evidence and targeting, not inflated claims.

Understand

- Every claim must map to course/project evidence.
- Preferred requirements are not automatically hard filters; explicit must-have constraints matter.
- Keep qualification, employment, location and experience statements factual.

Try / inspect

evidence -> claim -> verification -> publish

Common failure

Inflating experience/title, copying generic text or applying blindly without tracking reasons.

Your turn - independent proof

Audit P1/P2/P3/Capstone for README, architecture, reproduction, tests, failure/recovery and truthful claims.

Move on when

The artifact/decision can be defended line by line and revised from evidence.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L03 - LinkedIn/professional profile without overclaiming

Learning objective

Produce a truthful, evidence-backed job artifact or targeting decision.

Why this matters in Data Engineering

Job-launch quality depends on defensible evidence and targeting, not inflated claims.

Understand

- Every claim must map to course/project evidence.
- Preferred requirements are not automatically hard filters; explicit must-have constraints matter.
- Keep qualification, employment, location and experience statements factual.

Try / inspect

evidence -> claim -> verification -> publish

Common failure

Inflating experience/title, copying generic text or applying blindly without tracking reasons.

Your turn - independent proof

Write headline/About/featured-project plan and delete any statement you cannot defend.

Move on when

The artifact/decision can be defended line by line and revised from evidence.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L04 - Degree-aware job targeting: apply/skip rules

Learning objective

Produce a truthful, evidence-backed job artifact or targeting decision.

Why this matters in Data Engineering

Job-launch quality depends on defensible evidence and targeting, not inflated claims.

Understand

- Every claim must map to course/project evidence.
- Preferred requirements are not automatically hard filters; explicit must-have constraints matter.
- Keep qualification, employment, location and experience statements factual.

Try / inspect

evidence -> claim -> verification -> publish

Common failure

Inflating experience/title, copying generic text or applying blindly without tracking reasons.

Your turn - independent proof

Classify 15 fresh job descriptions as APPLY/MAYBE/SKIP and record the exact phrase that caused each decision.

Move on when

The artifact/decision can be defended line by line and revised from evidence.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L05 - SQL interview drills

Learning objective

Perform a timed unseen technical task and diagnose misses.

Why this matters in Data Engineering

Interview readiness requires recall and reasoning under time, not recognition of reviewed answers.

Understand

- Save first attempt before review.
- Correctness/grain/edge cases beat cleverness.
- Classify misses as syntax, logic, data grain, validation, debugging or time pressure.

Try / inspect

attempt -> save -> validate -> classify -> repair -> fresh retest

Common failure

Opening solutions early or memorizing a previous mock.

Your turn - independent proof

Solve a fresh 4-question SQL mock under time, stating grain and one validation per query.

Move on when

You can solve a different fresh prompt and explain your validation.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L06 - Python interview drills

Learning objective

Perform a timed unseen technical task and diagnose misses.

Why this matters in Data Engineering

Interview readiness requires recall and reasoning under time, not recognition of reviewed answers.

Understand

- Save first attempt before review.
- Correctness/grain/edge cases beat cleverness.
- Classify misses as syntax, logic, data grain, validation, debugging or time pressure.

Try / inspect

attempt -> save -> validate -> classify -> repair -> fresh retest

Common failure

Opening solutions early or memorizing a previous mock.

Your turn - independent proof

Implement a fresh 4-task Python mock with edge-case tests and explicit failure handling.

Move on when

You can solve a different fresh prompt and explain your validation.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L07 - Data-engineering concepts and troubleshooting drills

Learning objective

Explain engineering decisions, failures and recovery using your own projects or a fresh scenario.

Why this matters in Data Engineering

Interviewers test whether you understand the system, not whether the README sounds polished.

Understand

- State inputs/outputs and data grain.
- Explain one failure mode and recovery path.
- Use measured/observed evidence; do not invent production scale.

Try / inspect

problem -> design -> correctness -> failure -> recovery -> tradeoff

Common failure

Reciting tool definitions without connecting them to system behavior.

Your turn - independent proof

Answer 12 troubleshooting questions; prioritize safe diagnosis/recovery over memorized definitions.

Move on when

You can answer follow-ups without reading your notes.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L08 - Project walkthrough interview

Learning objective

Explain engineering decisions, failures and recovery using your own projects or a fresh scenario.

Why this matters in Data Engineering

Interviewers test whether you understand the system, not whether the README sounds polished.

Understand

- State inputs/outputs and data grain.
- Explain one failure mode and recovery path.
- Use measured/observed evidence; do not invent production scale.

Try / inspect

problem -> design -> correctness -> failure -> recovery -> tradeoff

Common failure

Reciting tool definitions without connecting them to system behavior.

Your turn - independent proof

Deliver a 5-minute project walkthrough and answer 8 unpredictable follow-ups about correctness, failure and tradeoffs.

Move on when

You can answer follow-ups without reading your notes.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L09 - Architecture/system-design defense for junior roles

Learning objective

Explain engineering decisions, failures and recovery using your own projects or a fresh scenario.

Why this matters in Data Engineering

Interviewers test whether you understand the system, not whether the README sounds polished.

Understand

- State inputs/outputs and data grain.
- Explain one failure mode and recovery path.
- Use measured/observed evidence; do not invent production scale.

Try / inspect

problem -> design -> correctness -> failure -> recovery -> tradeoff

Common failure

Reciting tool definitions without connecting them to system behavior.

Your turn - independent proof

Design a junior-level batch pipeline for a fresh scenario; state sources, storage, orchestration, quality, observability, recovery and tradeoffs.

Move on when

You can answer follow-ups without reading your notes.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L10 - Behavioral answers and career-transition story

Learning objective

Communicate truthful engineering evidence clearly and concisely.

Why this matters in Data Engineering

Strong technical work can be lost if the explanation is vague, inflated or unstructured.

Understand

- Use concrete facts and numbers from executed evidence.
- Separate what you did from what the team/tool does generally.
- State limitations and next actions directly.

Try / inspect

context -> action -> evidence -> result -> learning

Common failure

Long generic answers, fake confidence or jargon without a concrete example.

Your turn - independent proof

Answer 8 behavioral prompts using truthful Situation-Action-Result-Learning evidence.

Move on when

A listener can follow the answer and ask meaningful technical follow-ups.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L11 - Technical English: explain code, failures and decisions clearly

Learning objective

Communicate truthful engineering evidence clearly and concisely.

Why this matters in Data Engineering

Strong technical work can be lost if the explanation is vague, inflated or unstructured.

Understand

- Use concrete facts and numbers from executed evidence.
- Separate what you did from what the team/tool does generally.
- State limitations and next actions directly.

Try / inspect

context -> action -> evidence -> result -> learning

Common failure

Long generic answers, fake confidence or jargon without a concrete example.

Your turn - independent proof

Record five 60-90 second explanations of code/failure/decision; score clarity, structure, accuracy and filler words.

Move on when

A listener can follow the answer and ask meaningful technical follow-ups.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L12 - Application tracker, feedback loop and weekly repair cycle

Learning objective

Run a measurable application/interview feedback loop.

Why this matters in Data Engineering

Job search is an operational process that can be diagnosed and improved.

Understand

- Track application date, role, must-have fit, outcome and next action.
- Diagnose low screen rate vs low technical-pass rate differently.
- Refresh active market requirements from current job descriptions.

Try / inspect

applications -> screens -> tests -> interviews -> offers

Common failure

Changing the whole career plan after a few untracked rejections.

Your turn - independent proof

Maintain application tracker for one week and diagnose funnel bottleneck from counts instead of emotion.

Move on when

You can identify the current funnel bottleneck and one targeted repair.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L13 - Timed coding test: arrays, strings and hash-map problems

Learning objective

Perform a timed unseen technical task and diagnose misses.

Why this matters in Data Engineering

Interview readiness requires recall and reasoning under time, not recognition of reviewed answers.

Understand

- Save first attempt before review.
- Correctness/grain/edge cases beat cleverness.
- Classify misses as syntax, logic, data grain, validation, debugging or time pressure.

Try / inspect

attempt -> save -> validate -> classify -> repair -> fresh retest

Common failure

Opening solutions early or memorizing a previous mock.

Your turn - independent proof

Solve three timed coding problems and explain complexity plus edge cases.

Move on when

You can solve a different fresh prompt and explain your validation.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.

M19-L14 - Combined SQL + Python interview simulation

Learning objective

Perform a timed unseen technical task and diagnose misses.

Why this matters in Data Engineering

Interview readiness requires recall and reasoning under time, not recognition of reviewed answers.

Understand

- Save first attempt before review.
- Correctness/grain/edge cases beat cleverness.
- Classify misses as syntax, logic, data grain, validation, debugging or time pressure.

Try / inspect

attempt -> save -> validate -> classify -> repair -> fresh retest

Common failure

Opening solutions early or memorizing a previous mock.

Your turn - independent proof

Complete a 90-minute combined SQL + Python mock with saved first attempt and post-attempt error classification.

Move on when

You can solve a different fresh prompt and explain your validation.

Revision cadence: D+1 fresh retrieval; D+4 mixed task; D+10 repair scenario; D+30 interview retrieval.